

Yize Peng

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EDUCATION

Shanghai University of Finance and Economics

Sep. 2021 – Jun. 2025

Bachelor in Data Science and Big Data Technology

Core Courses: Mathematical Analysis, Data Structure, Algorithmic Design & Analysis, Advanced Operations Research

Columbia University in the city of New York | Exchange Program

Sep. 2023 – Jun. 2024

Core Courses: Ordinary Differential Equations, Analysis and Optimization, Operations Research in Public Policy, Healthcare Operations Management

University of Pennsylvania

Aug. 2025 – Dec. 2026

Master of Science in Social Policy and Data Analytics

GPA: 4.0/4.0

RESEARCH EXPERIENCES

Exploring the Best EOQ Under Different Circumstances

Jan. 2024 - Mar. 2024

MSU, under the guidance of Srinivas Talluri

- Responsible for the development and optimization of the EOQ Model, aiming to reduce overall cost in inventory management to respond uncertainty in demand.
- Proposed two innovation optimization algorithms, one is on the basis of the assumption of fundamental constants and the other a heuristic algorithm designed for real-time demand changes, which improved the performance of the model in complex tasks.
- Implemented numerical simulations to validate the effectiveness of algorithm, demonstrating the potential to minimize the cost by optimizing the ordering time.
- Incorporated a probability distribution model to deal with uncertain demands, and produced a comprehensive formula for calculating inventory costs.
- Considered the environmental factors into the EOQ Model to make inventory management more sustainable and be in line with corporate values in achieving green supply chain.
- Produced a paper that was accepted by the 2024 International Conference on Computer Science and Management Technology (ICCSMT 2024), and it will be published on the ACM ICPS (ISBN: 979-8-4007-0999-9)

Portfolio Optimization Based on LSTM for 5 Stocks

NYU, under the guidance of Miquel Noguer Alonso

Jan. 2023 - Mar. 2023

- Developed an optimization model based on LSTM to analyze the 5 representative stocks (Google, Microsoft, Nvidia, Amazon, Tesla) based on data collected from September 2022 to March 2023, and used deep learning techniques to predict stock performance.
- Used the LSTM model to learn the top 70% of the data, and predicts the bottom 30% of the data, then use the portfolio strategy for the highly accurate forecast results.
- Applied a portfolio optimization strategy with predicted data, and achieved the maximum return model and the minimum risk model on the effective frontier of Monte Carlo.
- The LSTM model is proved to be more effective by applying two models of the portfolio and combining two results with the S&P 500.
- Produced a paper and had it published to the 2023 International Conference on Management Research and Economic Development (DOI: 10.54254/2754-1169/25/20230504) and will be published in the Journal of Advances in Economics Management and Political Science (ISSN: 2754-1169)

INDUSTRY EXPERIENCES

Webacy | Data Analyst

Jul. 2024 – Sep. 2024

- Familiarized with different required risk tags, reviewed and grouped small contracts with audit reports and labeled their vulnerabilities on the contracts to flag the potential risks.
- Cleaned and managed the data with Excel and used Python to perform frequency analysis, correlation analysis, and cluster analysis to discover insights and identify common risks and study the correlation between them.
- Through frequency and correlation analysis, identified the common vulnerabilities and risks in smart contracts; visualized data in Tableau and organized the results into reports and delivered presentation to team members on the potential risks and optimization strategies.

Xi'an Yunjing Technology Inc. | Governance Big Data Analyst Intern

Jun. 2023 - Aug. 2023

- Engaged in the Xi'an Public Resources Transaction Process Monitoring and Optimization Project, and worked to process over 5,000 data including fields in water conservancy, transportation, housing construction to ensure the project meets the regulatory frameworks.
- Processed and optimized public resources transaction data with SQL and techniques such as data cleaning, normalization and missing value processing to eliminate data noise and errors.
- Used Tableau for data visualization and developed a real-time monitoring dashboard to identify any red flags during transactions.

PROJECT EXPERIENCE

Thousand-village Survey

Jun. 2022 - Sep. 2022

Team leader, Shanghai University of Finance and Economics

- Analyzed in-depth of the rural employment problem in the western China, identifying main challenges such as lack of skills, resources and talent insufficiency in Shaanxi Province, Guizhou Province and Ningxia Hui Autonomous Region through questionnaires and field trips.
- Processed and analyzed data from the Thousand-village Project in 2021 using descriptive statistics and regression analysis, identifying key factors in the employment problem after data cleaning, statistical analysis and data visualization with Excel and SPSS.
- Proposed an innovative solutions to implement the “corporate+rural residents” model in rural tourism in Guizhou Province, and analyzed the potential benefits and implementation methods of this model.

EXTRACURRICULAR ACTIVITIES

American College Student Mathematical Modeling Competition | Member

Feb. 2024

- Utilized models such as LSTM and Monte Carlo Simulation to analyze the development of wildlife trade

National College Student Mathematical Modeling Competition | Leader

Sep. 2022

- Led research on the bearing-only passive localization model in unmanned aerial vehicle based on mathematical model

Shanghai University of Finance and Economics Business Competition | Best Speaker

Feb. 2022

- Conducted market analysis for a local bubble tea brand
- Responsible for making “Porter’s Five Forces” analysis to analyze the market competition and bargaining power
- Ranked first out of 18 groups, and was rewarded the “Best Speaker”

SKILLS AND LANGUAGE

- Computer: C++, Python, R
- Standardized Tests: GRE (336: V: 166, Q: 170, AW: 4.5); TOEFL Total 105 (R: 30, L: 25, S:22, W:28)